

# STRUCTURALLY IMPOSSIBLE PROMPTS

(math & code)

MATH

$\cot(-540^\circ)$

CODE

`(1).startswith("1")`



**INSIDE THE MODEL** (one shared model)

✓ flags **IMPOSSIBLE** AUC **0.939**

$d_{\text{imp}}$  active

$\cos 0.087$

not read

safety-refusal  
gate  
(inactive)

$\cos 0.40$

partial align

behavior dir.  
 $d_{\text{struct,behav}}$

*the model answers anyway*

$\cot(-540^\circ) = 0$  ✗

True ✗

## CAUSAL INTERVENTION

steer  $d_{\text{imp}}$  during generation → “undefined — cannot compute” ✓

**gated  $\Delta G$  +33–52 pp** (4-anchor math/code grid)

16-model sweep: math/code footprint; fact weakest